

Mass-Bound Time Scales and Interferometry

Why Holometer-Type Experiments Find No Discreteness Signal –
a Derivation from $\tilde{T} \cdot m = 1$

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Chapter A

Mass-Bound Time Scales and Interferometry

Correction notes

Corrections from the calculation review (1 Oct. 2026). The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- Column $R_m = \hbar/(mc^2)$: unit m $\rightarrow$ s (neutron $7.006 \rightarrow 7.005 \times 10^{-25}$ s); lattice dispersion with the Compton length $\hbar/(mc)$: electron $\sim 10^{-44}/10^{-42} \rightarrow \sim 10^{-27}/4 \times 10^{-25}$, proton $\sim 10^{-51}/10^{-49} \rightarrow \sim 3 \times 10^{-34}/10^{-31}$.
- Gap between Compton frequency and interferometers: "at least 13 orders of magnitude" $\rightarrow$ at least 12 (QUEST: 1.5×10^{12}).
- Note: the K_{frak} shift of f_e is not a measurable spectral line (R135).

Abstract

From the FFGFT fundamental relation $\tilde{T} \cdot m = 1$ follow mass-bound Compton frequencies $f_m = mc^2/h$ as the only physically grounded eigenfrequencies of a discrete substrate **[K]**. For all stable particles these lie at least 12 orders of magnitude above the measurement frequencies of current interferometers (MHz range). (*Corrected on 1 Oct. 2026: previously "at least 13 orders of magnitude" – $f_e/f_c^{QUEST} \approx 1.5 \times 10^{12}$, i.e. $10^{12.2}$; only the Holometer, at 3.3×10^{13} , lies above 13.*) This yields a precise, derivable statement about interferometry experiments for the detection of space quantisation: not a design-induced blind spot but the mass-bound nature of the time scale $\tilde{T} = 1/m$ explains why Holometer-type experiments at their current sensitivity cannot find a discreteness signal **[K]**. A K_{frak} correction to the Compton frequency of $\Delta f_e/f_e = 100\xi_0 \approx 1.3\%$ is in principle spectroscopically accessible, but not interferometrically **[K]**. (*Note of 1 Oct. 2026: K_{frak} acts only on the SI conversion; the shift is not a measurable spectral line, see the section on the K_{frak} correction.*)

A.1 Background: Two Approaches to Space Quantisation

Experiments such as the Fermilab E-990 Holometer ($L = 40 \text{ m}$, $f_c = c/(2L) \approx 3.75 \text{ MHz}$) and the planned QUEST experiment ($L = 1.84 \text{ m}$, $f_c \approx 81.5 \text{ MHz}$) search for a quantisation signal of spacetime. The null results of these experiments have been interpreted from various quarters as evidence for or against specific discrete substrate models.

One concrete claim is that the interferometers are by design blind at exactly the frequency at which a discrete D4 lattice with lattice constant $a = \ell_P$ would resonate. This claim presupposes that the relevant substrate frequency is $f \sim c/(2L)$.

The FFGFT framework provides a more precise answer to the question of the relevant frequency scales — from first principles, without parameter fitting.

A.2 FFGFT Eigenfrequencies from $\tilde{T} \cdot m = 1$

The Mass-Bound Time Scale

[ASSUMPTION] In FFGFT, time is not an independent external quantity but is bound to mass by the fundamental relation $\tilde{T} \cdot m = 1$ (in natural units; in SI: $\tilde{T} \cdot m = \hbar/c^2$).

[K] Every massive particle therefore carries an intrinsic time scale:

$$\tilde{T}_m = \frac{\hbar}{mc^2}, \quad f_m = \frac{mc^2}{h}. \quad (\text{A.1})$$

This is the Compton frequency — not an additional postulate but a geometric consequence of the T^4 structure and the duality relation.

External Inputs

[ASSUMPTION] This document uses the following standard values as external inputs; they are declared here and not derived:

Quantity	Value	Source
m_e	$9.1094 \times 10^{-31} \text{ kg}$	PDG
m_μ	$1.8836 \times 10^{-28} \text{ kg}$	PDG
m_p	$1.6726 \times 10^{-27} \text{ kg}$	PDG
m_n	$1.6749 \times 10^{-27} \text{ kg}$	PDG
h	$6.6261 \times 10^{-34} \text{ J s}$	SI (derived from ξ)
c	$2.9979 \times 10^8 \text{ m/s}$	SI (derived from ξ)

Numerical Values

[K]

Compton frequencies of stable particles (SI)

Particle	$f_m = mc^2/h$	$R_m = \hbar/(mc^2)$
Electron	$1.2356 \times 10^{20} \text{ Hz}$	$1.288 \times 10^{-21} \text{ s}$
Muon	$2.5548 \times 10^{22} \text{ Hz}$	$6.230 \times 10^{-24} \text{ s}$
Proton	$2.2687 \times 10^{23} \text{ Hz}$	$7.015 \times 10^{-25} \text{ s}$
Neutron	$2.2719 \times 10^{23} \text{ Hz}$	$7.005 \times 10^{-25} \text{ s}$

(Corrected on 1 Oct. 2026: previously unit m in the column $R_m = \hbar/(mc^2)$ and neutron $7.006 \times 10^{-25} - \hbar/(mc^2)$ is a time ($J \cdot s/J = s$), neutron more precisely $7.0055 \times 10^{-25} s$; the associated length is the reduced Compton wavelength $\hbar/(mc)$: 3.862×10^{-13} , 1.868×10^{-15} , 2.103×10^{-16} , $2.100 \times 10^{-16} m$.)

Comparison with Interferometer Frequencies

[K]

$$\frac{f_e}{f_c^{\text{Holometer}}} = \frac{1.2356 \times 10^{20} \text{ Hz}}{3.747 \times 10^6 \text{ Hz}} \approx 3.3 \times 10^{13}, \quad (\text{A.2})$$

$$\frac{f_e}{f_c^{\text{QUEST}}} \approx 1.5 \times 10^{12}. \quad (\text{A.3})$$

The smallest mass-bound frequency (electron) thus lies 12 to 13 orders of magnitude above the measurement frequencies of current interferometers. (Corrected on 1 Oct. 2026: previously "13 orders of magnitude" – $\log_{10}(3.3 \times 10^{13}) = 13.5$, $\log_{10}(1.5 \times 10^{12}) = 12.2$.)

A.3 Lattice Dispersion Correction at the FFGFT Scale

[K] For a discrete substrate with lattice constant a , the lattice dispersion correction to linear dispersion is of order $(k \cdot a)^2$. At $k_1 = \pi/L$ this gives:

$$\left. \frac{\delta\omega}{\omega} \right|_{\text{lattice}} \sim \left(\frac{\pi a}{L} \right)^2. \quad (\text{A.4})$$

[K]

Lattice scale a	Holometer ($L = 40 \text{ m}$)	QUEST ($L = 1.84 \text{ m}$)
$\ell_p = 1.62 \times 10^{-35} \text{ m}$	$\sim 10^{-72}$	$\sim 10^{-70}$
$\hbar/(m_e c) = 3.86 \times 10^{-13} \text{ m}$	$\sim 10^{-27}$	$\sim 4 \times 10^{-25}$
$\hbar/(m_p c) = 2.10 \times 10^{-16} \text{ m}$	$\sim 3 \times 10^{-34}$	$\sim 10^{-31}$

(Corrected on 1 Oct. 2026: previously $a = R_{m_e} = 1.29 \times 10^{-21} \text{ m}$ with $\sim 10^{-44}/\sim 10^{-42}$ and $a = R_{m_p} = 7.0 \times 10^{-25} \text{ m}$ with $\sim 10^{-51}/\sim 10^{-49}$ – there the time value $\hbar/(mc^2)$ had been inserted as a length; with $a = \hbar/(mc)$: $(\pi a/40 \text{ m})^2 = 9.2 \times 10^{-28}$, $(\pi a/1.84 \text{ m})^2 = 4.3 \times 10^{-25}$ (electron), 2.7×10^{-34} and 1.3×10^{-31} (proton).)

Even at the Compton scale of the electron — which in the FFGFT framework is the physical fundamental scale, not ℓ_p — the lattice dispersion correction is $\sim 10^{-27}$, far beyond any measurable sensitivity. (Corrected on 1 Oct. 2026: previously $\sim 10^{-44}$ – see the table; the conclusion stands.)

A.4 The K_{frak} Correction and Its Measurability

[K] The fractal correction $K_{\text{frak}} = 1 - 100\xi_0$ with $\xi_0 = 4/30000$ acts exclusively on absolute SI values, not on the pullback isometry itself (Doc. 190, R96, R98). Applied to the Compton frequency of the electron:

$$\Delta f_e = f_e \cdot 100\xi_0 = f_e \cdot \frac{400}{30000} \approx 1.65 \times 10^{18} \text{ Hz}. \quad (\text{A.5})$$

This corresponds to a relative deviation of $100\xi_0 \approx 1.33\%$ from the Compton frequency. This correction is in principle spectroscopically accessible (precision spectroscopy at $\sim 10^{20}$ Hz, i.e. hard X-ray to gamma range), but not with a Michelson interferometer in the MHz range. (Note of 1 Oct. 2026: Δf_e is arithmetically correct, but by the statement above K_{frak} acts only on absolute SI values, i.e. on the SI anchoring (Doc. 190, R135); the shift is therefore not a directly measurable spectral line and not immediately accessible to spectroscopy.)

A.5 The Central Statement on Interferometry Experiments

Conclusion: Why Holometer-type experiments find no signal

[K] In FFGFT the relevant frequency scale for space quantisation is not $f \sim c/(2\ell_P)$ (universal Planck frequency) but $f_m = mc^2/h$ (mass-bound Compton frequency). This scale differs for every particle and lies for all stable particles at least 12 orders of magnitude above the measurement frequencies of current interferometers. (Corrected on 1 Oct. 2026: previously “at least 13” – QUEST: $f_e/f_c \approx 1.5 \times 10^{12}$.)

It follows directly: Holometer-type experiments find nothing because the time scale of discreteness is not universal but mass-bound — not because a design-induced blind spot of the instrument lies exactly at the substrate resonance frequency.

The statement is falsifiable: an interferometer signal at frequency f would require a massless or ultralight field with $m = hf/c^2$. For $f = 3.75$ MHz this gives $m \approx 2.8 \times 10^{-44}$ kg $\approx 3 \times 10^{-14} m_e$ — no known particle.

A.6 Relation to the D4 Lattice Argument

[K] The D4 lattice with lattice constant $a = \ell_P$ has 24 nearest neighbours (8 of type $(\pm 1, 0, 0, 0)$ and 16 of type $(\pm \frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2}, \pm \frac{1}{2})$). The dispersion relation of a scalar field on this lattice is:

$$\omega^2(\mathbf{k}) = \frac{c^2}{\ell_P^2} \sum_{\delta} (2 - 2 \cos(\mathbf{k} \cdot \delta \ell_P)). \quad (\text{A.6})$$

For $k_1 = \pi/L$ with $L = 40$ m one has $k_1 \ell_P \approx 1.3 \times 10^{-36}$, hence $k_1 \ell_P \ll 1$: the dispersion relation is exactly linear in this regime, identical to the continuum up to order $(k_1 \ell_P)^2 \sim 10^{-72}$. No special lattice resonance exists at $f_c = c/(2L)$ — the free spectral range $c/(2L)$ is an optical property of the interferometer, not a lattice property.

FFGFT replaces the D4 lattice argument with a stronger statement: not the lattice dispersion but the mass-bound time structure $\tilde{T} \cdot m = 1$ determines at which frequency a signal is to be expected.

Notation

[ASSUMPTION] Axiom or declared external input – not derived
[K] Core – derived from ξ and the assumptions, numerically verified
[B] Bridge – algebraically proved
[S] Sketch – plausible, not yet fully worked out
 Verification script: `pruef_335_paul_frequenzen.py` (5/5 assertions).
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